

1D Gaussian Distribution in Exponential Canonical form

$$p(x|\theta) = h(x) \exp(\eta(\theta)^T T(x) - A(\theta))$$

$$\begin{aligned} p(x|\mu, \sigma^2) &= \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) \\ &= \exp\left(-\frac{1}{2} \ln(2\pi\sigma^2)\right) \exp\left(-\frac{x^2}{2\sigma^2} - \frac{\mu^2}{2\sigma^2} + \frac{2\mu x}{2\sigma^2}\right) \\ &= \exp\left(\underbrace{\left(\frac{\mu}{\sigma^2} x + \left(-\frac{1}{2\sigma^2} x^2\right)\right)}_{\eta(\theta)^T T(x)} - \underbrace{\left(\frac{\mu^2}{2\sigma^2} + \frac{1}{2} \ln(2\pi\sigma^2)\right)}_{A(\theta)}\right) \end{aligned}$$

$$T(x) = \begin{bmatrix} x \\ x^2 \end{bmatrix} \quad \begin{array}{l} T(x) \text{ sufficiently} \\ \sum_i x_i, + \sum_i x_i^2 \end{array}$$

$$\eta(\theta) = \begin{bmatrix} \eta_1 \\ \eta_2 \end{bmatrix} = \begin{bmatrix} \frac{\mu}{\sigma^2} \\ -\frac{1}{2\sigma^2} \end{bmatrix} \quad \begin{array}{l} \text{Precision weighted} \\ \text{mean,} \\ \text{Negative half} \\ \text{precision. } (\tau = \frac{1}{\sigma^2}) \end{array}$$

$$A(\theta) = \frac{\mu^2}{2\sigma^2} + \frac{1}{2} \ln(2\pi\sigma^2)$$

In natural params:

$$A(\eta) = -\frac{\eta_1^2}{4\eta_2} + \frac{1}{2} \ln\left(-\frac{\pi}{\eta_2}\right)$$

Derivative w.r.t η yields the expected moment:

$$E[T(x)] = E\left[\begin{bmatrix} x \\ x^2 \end{bmatrix}\right]$$

$$h(x) = 1 \quad \text{or} \quad h(x) = (2\pi)^{-\frac{1}{2}} \quad \text{if (instead kept.)}$$

$$\eta(\theta)^T T(x) = \eta_1 x + \eta_2 x^2$$

Multiply Gaussian prior with Gaussian likelihood
Simply add their natural params linearly.

- Precision weighted mean update

- Kalman filter measurements update